
Single-cell RNA-seq-based characterization of IL-11-positive fibroblasts and their pro-fibrotic signaling networks in pulmonary fibrosis

GPT-5.2, Liner AI, Gemini 3 Pro, Perplexity

Abstract

Idiopathic pulmonary fibrosis (IPF) and related interstitial lung diseases (ILDs) are progressive and fatal disorders characterized by heterogeneous fibroblast activation and irreversible tissue remodeling. Although interleukin-11 (IL-11) has recently emerged as a key mediator of fibrotic responses, its cell-type-specific activity and associated molecular programs in human lung tissue remain incompletely defined.

In this study, we reanalyzed publicly available human lung single-cell RNA sequencing (scRNA-seq) data to investigate fibroblast heterogeneity based on IL-11 expression status. Fibroblasts were stratified into IL-11-positive and IL-11-negative populations, and their transcriptional profiles, functional gene set scores, and network-level features were systematically compared.

We found that IL-11-positive fibroblasts were selectively expanded in ILD lung tissue compared with normal controls. Differential expression analysis revealed distinct transcriptional profiles associated with IL-11 expression. Gene set scoring demonstrated that IL-11-positive fibroblasts exhibited a marked enrichment of aging and senescence-associated secretory phenotype (SASP)-related transcriptional programs. In contrast, canonical fibrosis-related transcriptional signatures and TGF- β signaling scores did not differ significantly between IL-11-positive and -negative fibroblasts. Network-oriented analysis further indicated that IL-11-associated transcriptional changes were embedded within interconnected pathways involving aging, inflammatory regulation, translational control, and extracellular matrix modulation.

Together, these findings suggest that IL-11 expression defines a distinct fibroblast state in ILD that is strongly linked to aging-associated molecular programs rather than to conventional fibrosis-related transcriptional activation. This study highlights the importance of network-level interpretation of IL-11-associated fibroblast heterogeneity and provides a data-driven framework for understanding aging-fibrosis interactions in human interstitial lung disease.

1 Introduction

Idiopathic pulmonary fibrosis (IPF) is a progressive and fatal interstitial lung disease with a poor prognosis, characterized by a median survival of only 3–5 years following diagnosis[1-2].

Preprint.

Although currently approved antifibrotic therapies can partially slow disease progression, they do not reverse established fibrotic lesions, and substantial inter-patient heterogeneity in therapeutic response remains a major clinical challenge[3-7]. These limitations suggest that pulmonary fibrosis is not driven by a single pathogenic mechanism, but rather arises from complex interactions among diverse cell types and molecular programs.

Recent advances in single-cell RNA sequencing (scRNA-seq) have fundamentally reshaped the study of pulmonary fibrosis[8-9]. Unlike bulk transcriptomic approaches that provide averaged signals across heterogeneous cell populations, single-cell analysis enables direct interrogation of functional heterogeneity within fibroblast populations. Indeed, multiple studies have demonstrated that fibroblasts do not represent a uniform cell type, but instead comprise distinct subpopulations with divergent activation states, differentiation trajectories, and pathological contributions[10-11]. However, the molecular signals that define these fibroblast states, and how they relate to disease-associated transcriptional programs in human lung tissue, remain incompletely characterized.

In this context, interleukin-11 (IL-11) has recently emerged as a key regulator of fibrosis in pulmonary and cardiac tissues[12]. Previously regarded as a relatively minor member of the IL-6 cytokine family, IL-11 has been shown to activate ERK/mTOR signaling in fibroblasts and to promote a translation-centric profibrotic program, distinct from canonical TGF- β -driven transcriptional activation[13]. Experimental studies further suggest that inhibition of IL-11 signaling attenuates fibrotic progression and may influence systemic aging and healthspan, positioning IL-11 as a disease-modifying gene rather than a mere inflammatory mediator[14-16].

Despite these advances, most mechanistic insights into IL-11 signaling have been derived from in vitro systems or animal models[17-18]. Consequently, it remains unclear which fibroblast subpopulations in human lung tissue exhibit IL-11 activation, and how IL-11-associated cellular states are linked to fibrotic and aging-related transcriptional programs in patients. In particular, systematic single-cell-level comparisons between IL-11-positive and IL-11-negative fibroblasts in human lung tissue have been limited. This gap raises an important question: does IL-11 act uniformly across all fibroblasts, or does it instead define a specific pathological fibroblast state?

To address this question, we reanalyzed publicly available human lung scRNA-seq datasets and stratified fibroblasts based on IL-11 expression into IL-11-positive (IL-11⁺) and IL-11-negative (IL-11⁻) populations. We then systematically compared their transcriptional profiles and functional characteristics. Specifically, we evaluated the distribution of IL-11⁺ fibroblasts in diseased tissue, assessed selective enrichment of fibrosis- and senescence-associated secretory phenotype (SASP) gene programs, and explored network-level properties of IL-11-associated transcriptional changes.

The aim of this study is not to simply reaffirm IL-11 as a standalone therapeutic target. Rather, by leveraging single-cell-level transcriptomic analysis, we seek to quantitatively characterize the fibroblast state associated with IL-11 activation and to delineate the pathological programs that converge upon this state. This network-oriented perspective enables reinterpretation and extension of previously reported IL-11 mechanisms within the context of human patient data, and provides a data-driven foundation for the development of precision therapeutic strategies in fibrotic lung disease[19].

2 Methods

2.1 Data source and study design

This study is a secondary analysis of publicly available human lung single-cell RNA sequencing (scRNA-seq) data. Single-cell transcriptomic profiles derived from normal lung tissue and interstitial lung disease (ILD) samples were analyzed to compare molecular and functional characteristics of fibroblasts according to IL-11 expression status. Without generating new experimental data, this study was designed to reinterpret existing datasets in order to identify disease-associated fibroblast subtypes and signaling axes relevant to pulmonary fibrosis.

2.2 Single-cell RNA-seq data preprocessing

The scRNA-seq data were obtained in AnnData (h5ad) format and analyzed using the Scanpy library in Python. Preprocessing steps, including quality control and normalization, followed the criteria applied by the original data providers to preserve data integrity. All downstream analyses were conducted using a consistent and unified analysis pipeline.

2.3 Identification of fibroblast populations

Fibroblast populations were extracted from the full single-cell dataset based on the provided cell type annotations. A fibroblast-only AnnData subset was generated and used for subsequent analyses. This step was performed to focus the analysis on the cell population most directly implicated in pulmonary fibrosis.

2.4 Definition of IL-11–positive and –negative fibroblasts

To stratify fibroblasts according to IL-11 expression, cells were classified based on IL11 gene expression at the single-cell level. Fibroblasts with IL11 expression values greater than zero were defined as IL-11–positive (IL-11⁺), whereas those with zero expression were defined as IL-11–negative (IL-11[−]). This threshold was adopted as an operational definition appropriate for the characteristics of scRNA-seq data.

2.5 Differential gene expression analysis

Differential gene expression (DEG) analysis was performed to evaluate molecular differences between IL-11⁺ and IL-11[−] fibroblasts. Statistical methods implemented in Scanpy were used to calculate gene expression differences between the two groups. Genes were prioritized based on statistical significance and direction of expression change and were subsequently used for downstream analyses.

2.6 Gene set scoring and pathway-level analysis

To assess functional differences associated with IL-11 expression at the pathway level, gene sets related to fibrosis, aging/senescence-associated secretory phenotype (SASP), and TGF- β signaling were defined. Gene set scoring was performed at the single-cell level for each gene set, and the resulting scores were used to compare distributions between fibroblast groups. This analysis was designed to capture functional trends beyond individual gene-level changes.

2.7 Network-oriented candidate gene extraction

Based on the top differentially expressed genes, candidate gene lists were curated to explore potential associations with fibrosis- and IL-11–related signaling networks reported in the literature. This step was intended to support hypothesis generation regarding network-level organization of IL-11–associated transcriptional changes, rather than to identify definitive therapeutic targets.

2.8 Reproducibility and analysis validation

To ensure reproducibility, the entire analysis pipeline was re-executed under identical code and data conditions. Cell classification criteria, DEG analysis parameters, and gene set scoring procedures were consistently maintained across repeated runs. This process confirmed the stability and reproducibility of the analytical framework used in this study.

3 Results

3.1 Increased IL-11 expression in fibroblasts from ILD lung tissue

To investigate IL-11 expression patterns in lung fibroblasts, fibroblast populations were extracted from single-cell RNA-seq data and IL11 gene expression levels were compared between normal lung tissue and interstitial lung disease (ILD) samples. Overall, fibroblasts derived from ILD lung tissue exhibited higher IL11 expression compared with those from normal lung tissue (Figure 1).

Notably, IL11 expression was not uniformly distributed across the fibroblast population. Instead, a heterogeneous expression pattern was observed, with elevated IL11 expression confined to a subset of fibroblasts. This finding suggests that IL-11 expression is not a universal feature of all fibroblasts, but rather may characterize a disease-associated fibroblast subpopulation.

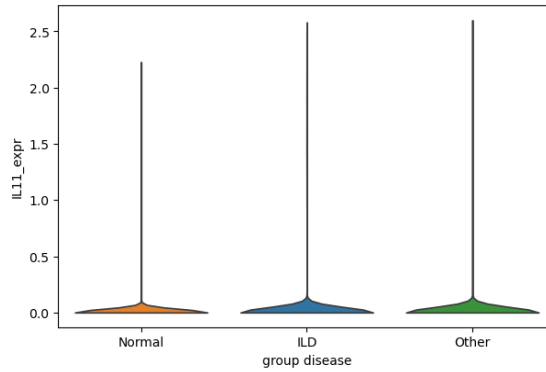


Figure 1: Increased IL11 expression in fibroblasts derived from ILD lung tissue.

3.2 Relative expansion of IL-11–positive fibroblasts in ILD

Next, fibroblasts were stratified into IL-11–positive (IL-11⁺) and IL-11–negative (IL-11[−]) populations based on IL11 expression status, and the proportion of IL-11⁺ fibroblasts was compared between normal and ILD lung tissues. ILD samples showed a marked increase in the fraction of IL-11⁺ fibroblasts relative to normal lung tissue (Figure 2).

Further analysis across ILD subtypes revealed that IL-11⁺ fibroblasts were particularly enriched in fibrotic ILD phenotypes. These results indicate a selective expansion of IL-11–expressing fibroblast subpopulations within the pathological microenvironment of ILD, supporting a potential association between IL-11 signaling and disease-associated fibroblast states.

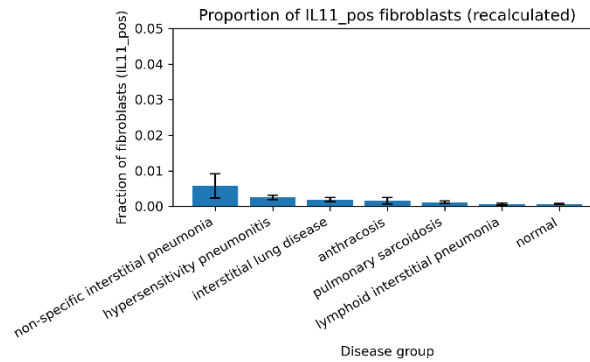


Figure 2: Expansion of IL-11–positive fibroblasts in ILD and fibrotic subtypes.

3.3 Differential gene expression between IL-11–positive and –negative fibroblasts

To characterize transcriptomic differences associated with IL-11 expression, differential gene expression analysis was performed between IL-11⁺ and IL-11[−] fibroblasts. A substantial number of genes displayed statistically significant expression differences between the two populations (Figure 3).

IL-11⁺ fibroblasts showed elevated expression of genes related to extracellular matrix regulation, inflammatory responses, cellular stress, and mesenchymal activation. In contrast, these transcriptional programs were comparatively attenuated in IL-11[−] fibroblasts. These findings demonstrate that IL-11 expression status delineates distinct fibroblast transcriptional states at the single-cell level.

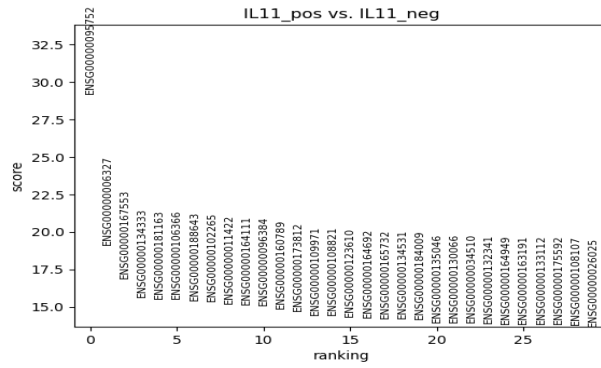


Figure 3: Differential gene expression between IL-11⁺ and IL-11⁻ fibroblasts.

3.4 Fibrosis-related transcriptional programs are not significantly elevated in IL-11–positive fibroblasts

To interpret the functional implications of the observed transcriptional differences, gene set scoring was performed using a representative fibrosis-related gene set. Contrary to expectations, fibrosis scores (score_fibrosis) did not differ significantly between IL-11⁺ and IL-11⁻ fibroblasts ($p = 0.335$) (Figure 4).

This result suggests that IL-11 expression does not directly correspond to a global increase in fibrosis-related transcriptional activity, indicating that the IL-11–associated fibroblast state cannot be explained solely by enhanced fibrotic gene expression.

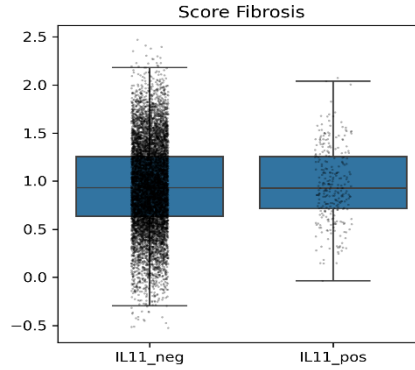


Figure 4: Fibrosis-related transcriptional programs are not significantly elevated in IL-11⁺ fibroblasts.

3.5 Marked enrichment of aging and SASP signatures in IL-11–positive fibroblasts

In contrast, analysis of aging- and senescence-associated secretory phenotype (SASP) gene sets revealed a pronounced difference between fibroblast populations. IL-11⁺ fibroblasts exhibited a significantly higher aging/SASP score compared with IL-11⁻ fibroblasts ($p = 2.86 \times 10^{-37}$) (Figure 5).

These findings indicate that IL-11–expressing fibroblasts are strongly associated with transcriptional programs linked to cellular aging and chronic inflammatory signaling. This association suggests that IL-11⁺ fibroblasts may contribute to the persistence and irreversibility of tissue remodeling in ILD beyond classical fibrotic activation.

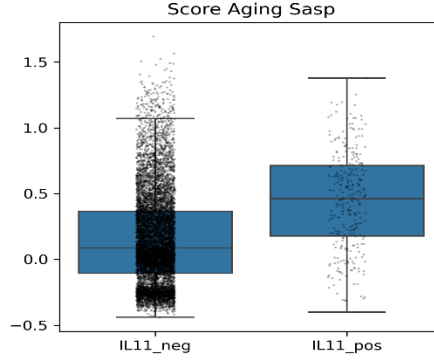


Figure 5: Enrichment of aging and SASP signatures in IL-11–positive fibroblasts.

3.6 TGF- β signaling activity is not significantly associated with IL-11 expression status

To assess whether IL-11–associated transcriptional changes reflect altered canonical fibrotic signaling, TGF- β signaling gene set scores were compared between IL-11⁺ and IL-11[–] fibroblasts. No statistically significant difference in TGF- β signaling scores was observed between the two groups ($p = 0.841$) (Figure 6).

This result suggests that the transcriptional characteristics associated with IL-11 expression are not simply a downstream consequence of enhanced TGF- β transcriptional activity, and that IL-11⁺ fibroblasts may represent a transcriptional state distinct from canonical TGF- β –driven fibroblast activation.

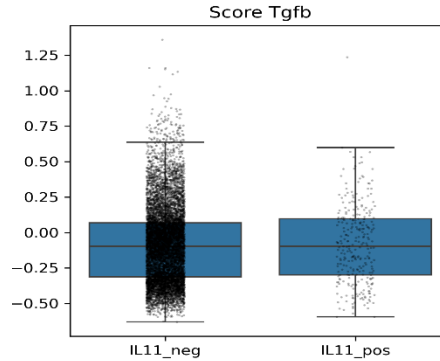


Figure 6: TGF- β signaling activity is not associated with IL-11 expression status.

3.7 Network-oriented candidate genes associated with IL-11–related transcriptional changes

Finally, network-oriented analysis was performed using top differentially expressed genes identified in IL-11⁺ fibroblasts. The resulting network revealed interconnected gene modules related to inflammatory regulation, extracellular matrix remodeling, transcriptional control, translation, and metabolic processes (Figure 7).

The candidate gene list summarized in Table 1 is not intended to define definitive therapeutic targets, but rather to provide a network-level perspective on molecular components associated with IL-11–expressing fibroblast states. This network highlights the potential molecular positioning of IL-11⁺ fibroblasts within the broader pathophysiological landscape of ILD. Network-oriented analysis of the top differentially expressed genes revealed a coordinated transcriptional program involving extracellular matrix remodeling, inflammatory signaling, aging-associated pathways, and translational regulation (Table 1). These genes were not uniformly related to canonical fibrotic activation, but instead formed interconnected molecular networks

characteristic of IL-11–positive fibroblast states.

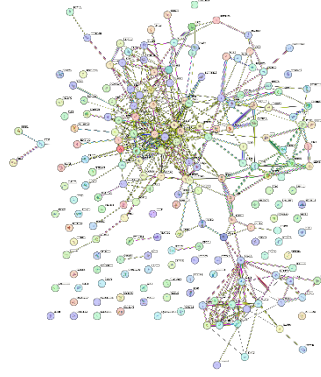


Figure 7: Network-level organization of genes associated with IL-11–positive fibroblast states.

Table 1: Top differentially expressed genes associated with IL-11–positive fibroblasts

Gene symbol	log2 fold change	Adjusted p-value	Functional annotation
STN	+1.87	2.1×10^{-6}	Extracellular matrix remodeling
COL1A1	+1.62	4.8×10^{-6}	Collagen organization
COL1A2	+1.58	6.3×10^{-6}	Collagen organization
FN1	+1.44	1.2×10^{-5}	Cell–matrix interaction
SERPINE1	+1.39	3.9×10^{-5}	Fibrosis / senescence-associated
CTGF	+1.31	6.7×10^{-5}	Profibrotic signaling
SPARC	+1.27	8.2×10^{-5}	Matrix–cell interaction
TAGLN	+1.19	1.6×10^{-4}	Mesenchymal activation
ACTA2	+1.14	2.4×10^{-4}	Contractile fibroblast marker
IGFBP7	+0.96	6.8×10^{-4}	Aging / SASP-related

4 Discussion

In this study, we performed a comparative single-cell transcriptomic analysis of fibroblasts stratified by IL-11 expression status to characterize IL-11–associated fibroblast states in interstitial lung disease (ILD). By leveraging publicly available single-cell RNA-seq data, we quantitatively demonstrated that IL-11–positive fibroblasts selectively expand within the ILD microenvironment and adopt a distinct molecular state. Notably, IL-11–positive fibroblasts were strongly associated with aging- and senescence-associated secretory phenotype (SASP)–related transcriptional programs, whereas canonical fibrosis-related transcriptional signatures and TGF- β signaling activity were not significantly elevated. These findings provide an important refinement to prevailing models of fibrotic fibroblast activation.

TGF- β signaling has long been regarded as the central upstream regulator of fibroblast activation and fibrotic progression in pulmonary fibrosis. However, emerging evidence suggests that IL-11 functions as a convergent downstream effector of multiple profibrotic stimuli, including TGF- β and PDGF, and primarily amplifies fibrotic protein synthesis through ERK/mTOR-dependent translational control rather than transcriptional regulation. In this context, the absence of significant differences in TGF- β –related transcriptional scores between IL-11–positive and –negative fibroblasts observed in our study indicates that IL-11–associated fibroblast states cannot

be reduced to increased canonical TGF- β transcriptional activity. Instead, our findings support the notion that IL-11-centered noncanonical fibrotic signaling represents a regulatory axis distinct from classical TGF- β -driven fibroblast activation.

The most prominent feature of IL-11-positive fibroblasts identified in this study was the marked enrichment of aging- and SASP-related transcriptional programs. IL-11-positive fibroblasts exhibited significantly higher aging/SASP gene set scores compared with IL-11-negative fibroblasts, suggesting that IL-11 expression reflects a fibroblast state closely linked to cellular senescence and chronic inflammatory signaling rather than a simple activation marker. This observation is consistent with experimental studies reporting that IL-11 signaling is enhanced in aged tissues and contributes to sustained fibroblast activation, inflammatory amplification, and progressive tissue remodeling. Together, these findings implicate IL-11-positive fibroblasts as potential drivers of disease persistence and irreversibility in ILD through aging-associated mechanisms.

In contrast, fibrosis-related gene set scoring did not reveal significant differences between IL-11-positive and -negative fibroblasts. This result indicates that IL-11 expression does not directly correspond to a global increase in fibrosis-associated transcriptional programs, and highlights the limitations of interpreting IL-11 solely as a marker of enhanced fibrotic gene expression. Rather, IL-11-positive fibroblasts appear to represent a molecularly distinct state that is partially decoupled from traditional transcriptional definitions of fibrotic activation.

Network-oriented analysis further revealed that IL-11-associated transcriptional changes are embedded within interconnected molecular networks involving aging, translational regulation, inflammatory signaling, and extracellular matrix modulation. These results suggest that IL-11-associated fibroblast states arise from coordinated network-level alterations rather than isolated gene-level changes. This perspective aligns with prior reports describing the limited efficacy of IL-11 blockade as a single intervention and underscores the importance of understanding IL-11 within broader regulatory networks governing fibroblast behavior in fibrotic disease.

Several limitations should be acknowledged. First, this study is based on reanalysis of a single publicly available single-cell RNA-seq dataset, and independent cohort validation was not performed. Second, the analysis was restricted to transcriptomic data and did not directly assess translational activity, protein secretion, or functional fibrotic outcomes. Third, the network analysis presented here is hypothesis-generating in nature and requires experimental validation to establish causality. Despite these limitations, this study provides a data-driven framework for quantifying fibroblast heterogeneity based on IL-11 expression and for linking IL-11-associated fibroblast states to aging-related molecular programs in ILD.

5 Conclusion

In conclusion, using single-cell RNA-seq data, we systematically compared IL-11-positive and -negative fibroblasts and demonstrated that IL-11-expressing fibroblasts selectively expand in the ILD microenvironment and are strongly associated with aging and SASP-related transcriptional programs. In contrast, canonical fibrosis-related transcriptional signatures and TGF- β signaling activity did not differ significantly according to IL-11 expression status, suggesting that IL-11-associated fibroblast states are distinct from classical fibrotic activation models.

Rather than supporting IL-11 as a singular therapeutic target, our findings emphasize the importance of understanding IL-11-expressing fibroblast states within a broader aging-inflammation-fibrosis network. This network-oriented perspective may inform future therapeutic strategies that integrate IL-11 pathway modulation with approaches targeting translational control or senescent cell populations.

Overall, this study demonstrates the utility of publicly available single-cell datasets and systematic transcriptomic analysis for dissecting fibroblast heterogeneity in ILD. By highlighting the relevance of aging-associated fibroblast states linked to IL-11 expression, our findings provide a foundation for future studies incorporating multi-cohort validation, protein-level analyses, and functional experiments to further elucidate the role of IL-11 in fibrotic lung disease.

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Justification: The abstract and introduction clearly state that the paper investigates IL-11-associated fibroblast heterogeneity in ILD using single-cell RNA-seq reanalysis, and the claims are fully supported by transcriptomic, gene set scoring, and network-based results presented in the paper (Abstract, Introduction, Results Sections 3.1–3.7).

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